A. What was changed in qwen3-tts.cpp (blue, marked +)

C interface and pipeline
originally one sentence at a time
+ 16 sentences a pass, refilled as they finish; + 12 predictor steps, a take check, all 15 for a failed take

Text tokenizer
byte-pair encoding
unchanged

Speaker encoder
run once per voice
unchanged

Transformers
talker + predictor
+ passes, 8-bit kernel
+ one attention for all
+ graphs kept

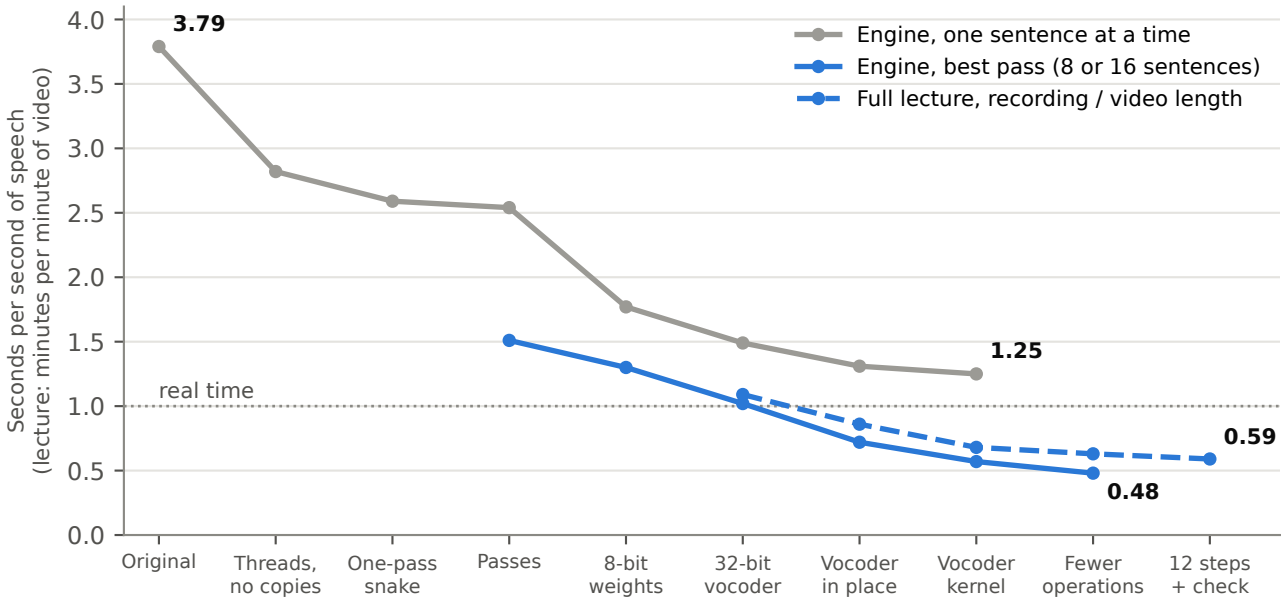
Vocoder
+ 32-bit, read in place
+ convolution kernel
+ 8-bit or GPU options

GGUF loader: 16- or 8-bit weights
+ the 8-bit model, mapped as it is; one conversion at load

GGML: a graph built, computed and reset for every step
+ the thread count applied; one thread pool; tinyBLAS gated to the vocoder

GPU and other accelerators
tried first by the original engine

CPU
was the fallback;
now the target

+ built on each computer for its processor: AVX2, AVX-VNNI, hybrid cores, NEON

B. What it bought on a 15-watt, 6-core laptop processor without a GPU